

1. Prove or disprove the following properties of divisibility:

- If $a \mid b$ and $b \mid c$ then $a \mid c$.
- If $x \neq 0$ and $ax \mid bx$ then $a \mid b$.
- If $a \mid b$ and $a \mid c$ then $a \mid (bx + cy)$ for any integers x, y .
- If $a \mid bc$ then $a \mid b$ or $a \mid c$.

Solution:

- True. If $a \mid b$ and $b \mid c$, then $b = ak$ and $c = b\ell$ for some $k, \ell \in \mathbb{Z}$. Hence $c = ak\ell$ and $a \mid c$.
- True. If $ax \mid bx$, then $bx = (ax)k$ for some $k \in \mathbb{Z}$. Since $x \neq 0$, dividing by x gives $b = ak$, so $a \mid b$.
- True. If $b = ak$ and $c = a\ell$, then $bx + cy = a(kx + \ell y)$, so $a \mid (bx + cy)$.
- False. For example, $6 \mid (2 \cdot 3)$, but 6 does not divide either 2 or 3.

2. Suppose that $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$. Prove or disprove the following:

- If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then $a + b \equiv c + d \pmod{m}$.
- If $a \equiv c \pmod{m}$ and $b \equiv d \pmod{m}$ then $ab \equiv cd \pmod{m}$.
- If $x \not\equiv 0 \pmod{m}$ and $ax \equiv bx \pmod{m}$ then $a \equiv b \pmod{m}$.

Solution: Recall $a \equiv c \pmod{m}$ means $m \mid (a - c)$.

- True. If $m \mid (a - c)$ and $m \mid (b - d)$ then

$$m \mid [(a - c) + (b - d)] = (a + b) - (c + d),$$

$$\text{so } a + b \equiv c + d \pmod{m}.$$

- True. Write $a = c + mk$ and $b = d + m\ell$. Then

$$ab = (c + mk)(d + m\ell) = cd + m(\cdots),$$

$$\text{so } ab \equiv cd \pmod{m}.$$

- False in general. For example, let $m = 6$, $x = 2$, $a = 1$, $b = 4$. Then

$$ax = 2 \equiv 8 = bx \pmod{6},$$

$$\text{but } 1 \not\equiv 4 \pmod{6}.$$

3. Prove that if $a \mid n$ then $2^a - 1 \mid 2^n - 1$ using modular arithmetic. Fun fact below.¹

¹A consequence of this is that $2^n - 1$ is prime only if n is prime. These numbers are called *Mersenne primes*. It is still an open problem whether there are infinitely many Mersenne primes.

Solution: If $a \mid n$, write $n = ak$. Then

$$2^n - 1 = 2^{ak} - 1 = (2^a)^k - 1$$

Note that $2^a \equiv 1$ modulo $2^a - 1$. Hence:

$$(2^a)^k - 1 \equiv 1^k - 1 \equiv 0 \pmod{2^a - 1}$$

as desired.

4. Prove that $9 \mid n$ if and only if the sum of the digits of n is divisible by 9. For example, $9 \mid 162$ and $1 + 6 + 2 = 9$. Hint: start by trying to write an equation relating n to its digits.

Solution: Write $n = \sum_{i=0}^r d_i 10^i$, where d_i are the digits of n . Since $10 \equiv 1 \pmod{9}$, we have $10^i \equiv 1 \pmod{9}$ for all i . Thus

$$n \equiv \sum_{i=0}^r d_i \pmod{9},$$

so $9 \mid n$ if and only if $9 \mid \sum d_i$.

5. (Fun, unrelated, and challenging problem) Suppose I secretly select a number x between 1 and 100 inclusive. We will then play the following game: each round you make a guess y and I will tell you whether $y \leq x$ or $y > x$. The game ends after you overestimate x two times. Your goal is to determine x . What is the fewest number of rounds it takes to guarantee you find the correct x , and what is your guessing strategy?

Solution: This problem is a version of the classic two-egg problem.

Suppose your first guess is y_1 . If $y_1 > x$, then you know $1 \leq x \leq y_1 - 1$. Since you are allowed only one more overestimate, you must then guess sequentially $1, 2, 3, \dots$ until reaching x . In the worst case ($x = y_1 - 1$), this requires y_1 total guesses (counting the guess y_1 itself).

If instead $y_1 \leq x$, you make a second guess $y_2 > y_1$. If $y_2 > x$, then $x \in \{y_1, \dots, y_2 - 1\}$ and again you must search linearly. In the worst case this requires $y_2 - y_1 + 1$ total guesses.

To minimize the worst-case number of guesses, we balance these cases by choosing

$$y_2 - y_1 + 1 = y_1,$$

so that the second interval is one smaller than the first. Continuing in this way, the sizes of the intervals decrease by 1 each time. If y_1 is too small, these intervals won't cover all of $\{1, \dots, 100\}$. For example, if y_1 is 3 then our guesses will be $y_1 = 3, y_2 = 5, y_3 = 6$ and we fail to reach 100. In general, we see our final guess will be:

$$y_1 + (y_1 - 1) + (y_1 - 2) + \dots + 1 = \frac{y_1(y_1 + 1)}{2}$$

We want to choose the smallest y_1 so that this procedure covers all of $\{1, \dots, 100\}$. Using the above formula, we see this is precisely $y_1 = 14$. Hence we can determine x in at most 14 rounds.